# mnirs v0.4.2.2

**Muscle Near-Infrared Spectroscopy Processing and Analysis**

Read, process, and analyse data from muscle near-infrared spectroscopy (mNIRS) devices. Import raw data from .csv or .xls(x) files and return a time-series data frame with metadata. Includes standardised methods for cleaning, filtering, and pre-processing mNIRS data for subsequent analysis, with a custom plot theme and colour palette.

## Workflow

```
├── read_mnirs()
│   ├── create_mnirs_data()
│   └── example_mnirs()
├── resample_mnirs()
├── replace_mnirs()
│   ├── replace_invalid()
│   ├── replace_outliers()
│   └── replace_missing()
├── filter_mnirs()
│   ├── filter_ma()
│   └── filter_butter()
├── shift_mnirs()
├── rescale_mnirs()
├── plot.mnirs()
│   ├── theme_mnirs()
│   ├── palette_mnirs()
│   ├── scale_colour_mnirs()
│   ├── scale_fill_mnirs()
│   ├── format_hmmss()
│   └── breaks_timespan()
└── extract_intervals()
```

## Read

`read_mnirs()` — Import mnirs data from .csv or .xls(x)

**Package-level functions**

`example_mnirs()` — Paths to an included example file

`create_mnirs_data()` — Add *"mnirs"* class metadata to data

## Resample

`resample_mnirs()` — Resample to a regular time grid

## Clean

`replace_mnirs()` — Replace outliers, invalid, and missing data

**Vector-level functions**

`replace_outliers()` — Replace local outliers

`replace_invalid()` — Replace specified invalid values

`replace_missing()` — Interpolate over *NA* values

## Filter

`filter_mnirs()` — Apply digital filter to mNIRS channels

**Vector-level functions**

`filter_butter()` — Apply a Butterworth digital filter

`filter_ma()` — Apply centred moving average smoothing

## Transform

`shift_mnirs()` — Shift channels to a reference value

`rescale_mnirs()` — Rescale channels data range

## Plot

`plot()` — Plot a data frame of class *"mnirs"*

`theme_mnirs()` — ggplot2 theme for mnirs plots

`palette_mnirs()` — Access the mnirs colour palette

`scale_colour_mnirs()` — Colour scale with mnirs palette

`scale_fill_mnirs()` — Fill scale with mnirs palette

`format_hmmss()` — Plot axis in time units

`breaks_timespan()` — Plot axis with time breaks

## Extract

`extract_intervals()` — Detect events and extract intervals